An Analysis of Model-Based Interval Estimation for Markov Decision Processes An Experimental Reproduction and Extension Study

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Abstract

We reproduce and extend the experiments of Strehl and Littman (2008), which introduces Model-Based Interval Estimation (MBIE) and proves the first fully explicit PAC-MDP sample-complexity bound for a confidence-interval-based reinforcement learning algorithm. We implement four algorithms — MBIE, MBIE-EB, R-Max, and E^3 — and reproduce the paper's main benchmark results on RiverSwim and SixArms using 10 independent trials. We then run five extensions: learning curves, sensitivity to the visit threshold m, sensitivity to the confidence-interval parameters A and B, sensitivity to the discount factor γ, and a test on FrozenLake 8×8. Our results confirm the paper's central claim: confidence interval based methods outperform threshold-based methods in practice. On SixArms, MBIE-EB (≈ 8.3M) and MBIE (≈ 6.2M) collect over $3\times$ the reward of R-Max and E^3 (≈ 2.5M and ≈ 2.1M). We identify and fix one reproducibility issue: a bug in E^3's escape probability rollout that prevents exploration.

1. Introduction

The exploration–exploitation dilemma is central to reinforcement learning (RL): an agent acting in an unknown MDP must balance gathering new information with acting on what it already knows. Strehl and Littman (2008) address this formally through the PAC-MDP framework, providing the first fully explicit sample-complexity bound for MBIE. The central principle is *optimism in the face of uncertainty*: uncertain state-action pairs receive the most favourable values consistent with current data, naturally driving principled exploration.

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In this seminar project, we reproduce several of the original experiments and explore how these algorithms behave beyond the settings of the paper. Our contributions are as follows.

- Providing an intuitive explanation of MBIE, MBIE-EB, R-Max, and E^3, including their PAC-MDP guarantees.
- Reproducing Figures 2 and 3 from the paper on RiverSwim and SixArms.
- Implementing five extensions probing parameter sensitivity, learning dynamics, and generalisation to new environments.

Our findings converge to the paper's main insight: confidence interval based methods outperform threshold-based methods in practice. Our extensions reveal parameter sensitivity, learning dynamics, and the value of principled exploration beyond the paper's designed environments.

Organisation of the report. Section 2 reviews necessary background. Section 3 describes the four algorithms with PAC-MDP guarantees. Section 4 details environments and implementation. Section 5 presents the reproduction results. Section 6 covers five extensions. Section 7 discusses broader implications, and Section 8 concludes.

2. Background and Basic Concepts

2.1. Markov Decision Processes

A finite MDP is a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, T, R, \gamma \rangle$ where $\mathcal{S}$ is a finite state space, $\mathcal{A}$ a finite action space, $T(s'|s,a)$ the transition distribution, $R(s,a)$ the expected reward bounded in $[0, R_{\max}]$, and $\gamma \in [0,1)$ the discount factor. The agent occupies a single state $s \in \mathcal{S}$, selects an action $a \in \mathcal{A}$, receives reward $r \sim R(s,a)$, and transitions to $s' \sim T(s,a)$. This repeats indefinitely.

A *policy* π is any strategy for choosing actions. The discounted infinite-horizon value function of policy π is:

$$V^\pi(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,\middle|\, s_0 = s,\, \pi\right].$$

The optimal action-value function satisfies the Bellman equation:

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} T(s'|s,a) \max_{a'} Q^*(s',a').$$

Note that no policy can have value greater than $v_{\max} = R_{\max}/(1-\gamma)$. A policy π is ε-optimal if $V^\pi(s) \geq V^*(s) - \varepsilon$ for all $s \in \mathcal{S}$.

2.2. PAC-MDP Framework

Definition 2.1 (Sample complexity of exploration, Kakade 2003)**.** Let $c = (s_1, a_1, r_1, s_2, a_2, r_2, \ldots)$ be a path generated by executing algorithm A in MDP $\mathcal{M}$. For any fixed $\varepsilon > 0$, the *sample complexity of exploration* of A with respect to c is the number of timesteps t such that A's policy A_t is not ε-optimal from state s_t, i.e. $V^{A_t}(s_t) < V^*(s_t) - \varepsilon$.

Definition 2.2 (Efficient PAC-MDP, Strehl and Littman 2008)**.** An algorithm A is said to be an *efficient PAC-MDP* algorithm if, for any ε and δ, the per-step computational complexity and the sample complexity of A are each less than some polynomial in the relevant quantities ($|\mathcal{S}|$, $|\mathcal{A}|$, $1/\varepsilon$, $1/\delta$, $1/(1-\gamma)$), with probability at least $1-\delta$.

The PAC-MDP framework is more "online" than classical PAC learning: mistakes can happen at any time, and the agent need not know when it has found a near-optimal policy. The only requirement is that suboptimal behaviour occurs on at most polynomially many timesteps in total.

3. Core Algorithms

All four algorithms share the principle of *optimism in the face of uncertainty*: unknown state-action pairs are assigned the most favourable values consistent with current data, naturally driving principled exploration.

3.1. MBIE — Model-Based Interval Estimation

MBIE (Strehl and Littman, 2008) maintains empirical estimates $\hat{R}(s,a)$ and $\hat{T}(\cdot|s,a)$ along with confidence intervals on each. Figure 1 illustrates the complete loop.[1]

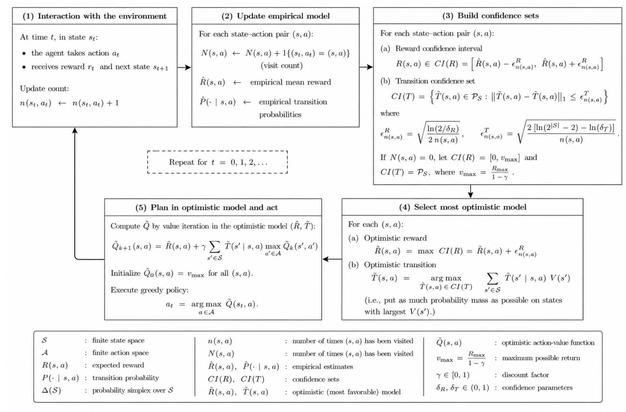

Figure 1: The MBIE loop: (1) interact with the environment; (2) update empirical model; (3) build confidence sets; (4) select the most optimistic model within the CIs; mbi (5) plan in the optimistic model and execute.

Reward confidence interval. After $n(s,a)$ experiences of (s,a), the empirical mean reward is $\hat{R}(s,a) = \frac{1}{n(s,a)} \sum_{i=1}^{n(s,a)} r[i]$. By the Hoeffding bound, with probability at least $1 - \delta_R$:

$$R(s,a) \in \mathrm{CI}(R) := \Big(\hat{R}(s,a) - \epsilon^R_{n(s,a)},\ \hat{R}(s,a) + \epsilon^R_{n(s,a)}\Big),$$

where $\epsilon^R_{n(s,a)} = \sqrt{\ln(2/\delta_R)\,/\,2n(s,a)}$.

Transition confidence interval. The empirical transition vector is $\hat{T}(s'|s,a) = n(s,a,s')/n(s,a)$. By Weissman et al. (2003), with probability at least $1 - \delta_T$, $T(s,a) \in \mathrm{CI}(T)$ where

$$\mathrm{CI}(T) := \left\{\tilde{T}(s,a) \in \mathcal{P}_{\mathcal{S}} \,\middle|\, \|\tilde{T}(s,a) - \hat{T}(s,a)\|_1 \leq \epsilon^T_{n(s,a)}\right\},$$

$\epsilon^T_{n(s,a)} = \sqrt{2\big[\ln(2^{|\mathcal{S}|} - 2) - \ln(\delta_T)\big]\,/\,m}$, and $\mathcal{P}_{\mathcal{S}}$ is the set of all probability distributions over $\mathcal{S}$.

Optimistic planning. On each timestep, MBIE solves the following system via value iteration:

$$\tilde{Q}(s,a) = \max_{\tilde{R} \in \mathrm{CI}(R)} \tilde{R}(s,a) + \max_{\tilde{T} \in \mathrm{CI}(T)} \gamma \sum_{s'} \tilde{T}(s'|s,a) \max_{a'} \tilde{Q}(s',a').$$

Unknown pairs ($n(s,a) = 0$) are initialised to $v_{\max}$. The inner maximisation over $\tilde{T}$ is solved by the greedy procedure of Strehl and Littman (2008, Section 3.1.5): probability mass is shifted toward the highest-value next state subject to the L_1 constraint. This update is a contraction mapping (Proposition 1), so value iteration converges to the unique fixed point $\tilde{Q}$.

[1]Diagram created with the assistance of ChatGPT (OpenAI, 2025).

PAC-MDP guarantee. With probability $\geq 1 - \delta$, MBIE acts suboptimally on at most (Theorem 1 of Strehl and Littman 2008):

$$\tilde{O}\left(\frac{|\mathcal{S}|^2|\mathcal{A}|}{\varepsilon^3(1-\gamma)^6}\right) \text{ timesteps.}$$

3.2. MBIE-EB — MBIE with Exploration Bonus

MBIE-EB replaces the full confidence interval optimisation with a scalar bonus added directly to the reward estimate:

$$\tilde{Q}(s,a) = \hat{R}(s,a) + \frac{\beta}{\sqrt{n(s,a)}} + \gamma \sum_{s'} \hat{T}(s'|s,a) \max_{a'} \tilde{Q}(s',a').$$

The bonus $\beta/\sqrt{n(s,a)}$ shrinks with visits, gradually replacing optimism with the empirical value. Unknown pairs ($n(s,a) = 0$) are initialised to $v_{\max}$.

The theoretical value of β that guarantees optimism (Lemma 7) is:

$$\beta = \frac{1}{1-\gamma}\sqrt{\frac{\ln(2|\mathcal{S}||\mathcal{A}|m/\delta)}{2}}.$$

MBIE-EB achieves the same PAC-MDP bound as MBIE (Theorem 2 of Strehl and Littman 2008) but requires only standard value iteration, making it simpler and cheaper to compute.

3.3. R-Max

R-Max (Brafman and Tennenholtz, 2002) applies optimism via a hard binary threshold:

- **Unknown pairs:** assigned fictitious reward $R_{\max}$ and a self-loop transition, making them maximally attractive in the internal model.
- **Known pairs:** use empirical reward $\hat{R}(s,a)$ and empirical transition $\hat{T}(s'|s,a)$.
- **Replanning:** value iteration is re-run only when a pair transitions from unknown to known, limiting computational cost.

The key insight is that the agent never needs to explicitly choose between exploration and exploitation. Acting greedily on the optimistic model either leads to a known, near-optimal region (exploitation) or to an unknown pair (exploration).

PAC-MDP guarantee. R-Max is PAC-MDP (Brafman and Tennenholtz, 2002). The argument has three steps: (1) after K_1 visits per pair, empirical estimates are accurate with high probability; (2) there are at most $|\mathcal{S}||\mathcal{A}|$ pairs to learn, each requiring K_1 visits, so total exploration is polynomially bounded; (3) once all pairs are known, the greedy policy is ε-optimal. R-Max provides a qualitative PAC bound, polynomial in $|\mathcal{S}|$, $|\mathcal{A}|$, $1/\varepsilon$, $1/\delta$, but without an explicit closed form. This contrasts with MBIE, whose bound $\tilde{O}(|\mathcal{S}|^2|\mathcal{A}|/\varepsilon^3(1-\gamma)^6)$ makes the cost of exploration directly quantifiable. Furthermore, R-Max's binary threshold prevents partial exploitation: a pair visited $m-1$ times contributes nothing to the model, whereas MBIE's CIs shrink continuously with each visit.

3.4. E^3 — Explicit Explore or Exploit

E^3 (Kearns and Singh, 2002) maintains **two separate models and policies** simultaneously:

- **Exploit model**: unknown pairs $\to$ reward 0, self-loop $\to$ policy *avoids* unknowns.
- **Explore model**: unknown pairs $\to v_{\max}$, self-loop $\to$ policy *seeks* unknowns.

At each step, E^3 estimates the *escape probability*: the probability that the explore policy reaches an unknown pair within H steps. If this exceeds a threshold τ, it follows the explore policy; otherwise it exploits.

PAC-MDP guarantee. E^3 was one of the first PAC-MDP algorithms (Kearns and Singh, 2002). The proof relies on the Explore-or-Exploit Lemma: for any set of known state-action pairs and any MDP, either there exists a policy whose T-step return is nearly optimal, or there exists a policy that reaches an unknown pair with probability $\geq \alpha/R_{\max}$. The total number of suboptimal steps is bounded polynomially in $|\mathcal{S}|$, $|\mathcal{A}|$, T, $1/\varepsilon$, $1/\delta$.

Unlike MBIE and R-Max, which are always implicitly optimistic, E^3 makes an **explicit binary decision** at every step, elegant in theory but brittle in practice, as discussed in Section 4.

4. Implementation

4.1. Environments

We reproduce the two benchmark MDPs from Section 6 of Strehl and Littman (2008), following Figure 1 of the paper exactly.

RiverSwim. 6 states arranged in a chain. The agent starts in state 1 or 2 with equal probability. Two actions are available: swim left (with the current) or swim right (against the current).

- **Left action**: always succeeds; reward 5 at state 0.
- **Right action**: moves forward with prob. 0.3, stays with prob. 0.7; reward 10,000 at state 5.

The expected reward at state 5 is $3{,}000 = 10{,}000 \times 0.3$. The challenge is that the agent must commit to a long sequence of rightward actions before discovering the large reward — naive exploration fails.

SixArms. 7 states: one hub (state 0) connected to 6 rooms (states 1–6). Each action at the hub pulls one arm, entering the corresponding room with some probability.

- **Arm probabilities**: [1.0, 0.15, 0.10, 0.05, 0.03, 0.01].
- **Room rewards**: [50, 133, 300, 800, 1,660, 6,000].

Higher-probability arms yield lower rewards. The agent must learn which arms are worth visiting despite having low success probability.

4.2. Parametrisation of the Confidence Intervals

Following Strehl and Littman (2008, Section 6), we parametrise the confidence intervals for MBIE as:

$$\epsilon^R_{n(s,a)} = A \cdot \frac{R_{\max}}{\sqrt{n(s,a)}}, \qquad \epsilon^T_{n(s,a)} = B \cdot \frac{1}{\sqrt{n(s,a)}},$$

where A and B are tunable scaling factors. For MBIE-EB, the exploration bonus is:

$$\frac{\beta}{\sqrt{n(s,a)}} = C \cdot \frac{R_{\max}}{\sqrt{n(s,a)}},$$

where $C = \beta / R_{\max}$. Parameter values used in reproduction follow the paper: $A = 0.3$, $B = 0$ (RiverSwim); $A = 0.3$, $B = 0.08$ (SixArms); $C = 0.4$ (RiverSwim); $C = 0.8$ (SixArms).

4.3. Computational Optimisations

Doubling replan schedule. Value iteration is expensive — running it at every timestep would require 5,000 replans per trial. We instead trigger replanning only at visit counts that are powers of 2 (i.e. $n(s,a) \in \{1, 2, 4, 8, \ldots\}$), reducing replanning calls from 5,000 to $\approx$200 per trial with negligible quality loss. This is valid because the model changes only when new data arrives, and the information gain per additional visit decreases rapidly.

Model size limit m. Following the paper, counts $n(s,a)$ and $n(s,a,s')$ are capped at m — data observed after the first m experiences of (s,a) is ignored. This bounds the total number of model updates to $|\mathcal{S}||\mathcal{A}|m$, limiting both computational and sample complexity. In the reproduction experiments we set $m = \infty$, to follow the algorithm's inputs specified by the paper.

4.4. Bug Fixes

E^3 escape probability. In a naive implementation of E^3, the escape probability rollout follows Q_{exploit} instead of Q_{explore}. The exploitation policy assigns $Q = 0$ to unknown pairs and loops on the best known state, causing $p_{\text{escape}} = 0$ and the agent never exploring. Paper Section 6 of Kearns and Singh (2002) explicitly states the rollout must follow the *exploration* policy. We fix this by running Monte Carlo rollouts under Q_{explore} to estimate the escape probability.

5. Reproducing the Paper Results

Reproducibility note. Strehl and Littman (2008) provide neither source code nor a complete description of their experimental setup. Parameter values are reported only as the result of "a broad search" (Section 6 of the paper), without specifying details such as the search grid, number of trials or random seed. Programming setup such as the programming language, libraries and hardware specifics were not mentioned. We therefore reconstructed all implementation details independently, choosing the missing parameters to best reproduce the qualitative behaviour shown in Figures 2 and 3 of the paper. The algorithms's input values given by the paper were: $A = 0.3$, $B_R = 0$, $B_S = 0.08$ (MBIE); $C_R = 0.4$, $C_S = 0.8$ (MBIE-EB); $m_R = 16$, $m_S = 6$ (R-Max); $m_R = 16$, $m_S = 4$, $thresh_R = 0.01$, $thresh_S = 0.09$ (E^3); $\gamma = 0.95$.

Our chosen experiment values: 5,000 steps, 10 independent trials.

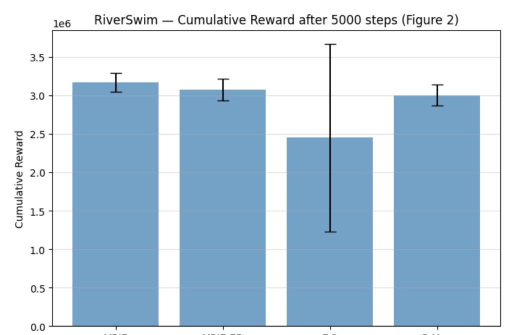

Figure 2: Cumulative reward on RiverSwim after 5,000 steps (reproduction of paper Fig. 2). Mean $\pm$ std, 10 trials.

RiverSwim (Figure 2). All four algorithms discover state 5 and collect $\approx$3M reward. The ordering MBIE ($\approx$3.17M) $>$ MBIE-EB ($\approx$3.08M) $>$ R-Max ($\approx$3.01M) $>$ E^3 ($\approx$2.45M) matches the paper. E^3 has a large standard deviation ($\approx$ 1.22M): one unlucky run gets trapped due to an unlucky batch of $m = 16$ rightward bounces that fail (probability $0.7^{16} \approx 0.3\%$).

Figure 3: Cumulative reward on SixArms after 5,000 steps (reproduction of paper Fig. 3). Mean ± std, 10 trials.

SixArms (Figure 3). MBIE-EB ($\approx$ 8.3M) $>$ MBIE ($\approx$ 6.2M) $\gg$ R-Max ($\approx$ 2.5M) $\approx$ E^3 ($\approx$ 2.1M), a 3$\times$ gap between CI-based and threshold-based methods. CI-based methods rule out inferior arms early, concentrating exploration on the highest-reward arms. Threshold-based methods must visit every arm m times regardless, wasting steps on clearly suboptimal actions. The high variance for MBIE-EB (std $\approx$4.9M) and E^3 (std $\approx$2.2M) reflects the bimodal nature of SixArms: runs either find the best arm quickly or waste the entire budget on suboptimal ones.

6. Extensions

Beyond reproducing the original experiments, we examine how the algorithms behave in settings not covered by the paper. These extensions test parameter sensitivity, learning dynamics, and generalisation to new environments.

6.1. Extension 1: Learning Curves

Bar charts show only the final cumulative reward and hide *when* each algorithm transitions from exploration to exploitation. Learning curves reveal this temporal structure.

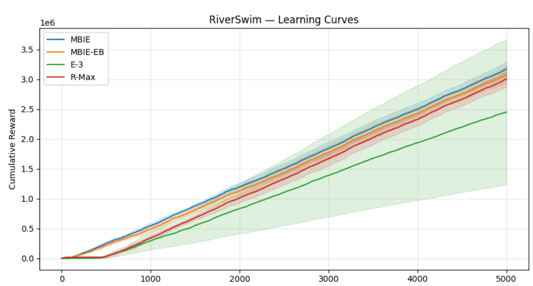

Figure 4: RiverSwim learning curves (mean ± std, 10 trials).

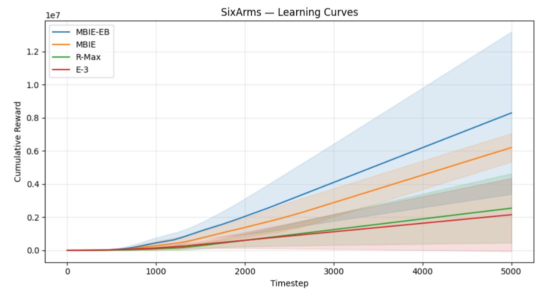

Figure 5: SixArms learning curves (mean ± std, 10 trials).

RiverSwim (Figure 4). MBIE and MBIE-EB begin collecting reward around timestep 500, with MBIE leading slightly throughout. R-Max follows a similar trajectory, converging to near-optimal behaviour around timestep 1,000. E^3's band is very wide: most runs perform similarly to the other algorithms, but one unlucky run never finds the high-reward state, inflating the standard deviation throughout.

SixArms (Figure 5). MBIE-EB leads throughout, rising steeply from $\approx$ 800 steps. MBIE follows a similar but lower trajectory. R-Max and E^3 grow much more slowly, threshold-based methods must visit every arm m times before exploiting, wasting the early budget on suboptimal arms. The gap between CI-based and threshold-based methods widens continuously over the 5,000 steps.

6.2. Extension 2: Sensitivity to m

We swept $m \in \{2^0, 2^1, \ldots, 2^7\}$ across all four algorithms (5 trials, 5,000 steps each). Theory (Theorem 1) gives $m \approx$ 200,000 for our parameters — far larger than the paper's $m = 16$, illustrating the typical gap between PAC worst-case bounds and practical performance.

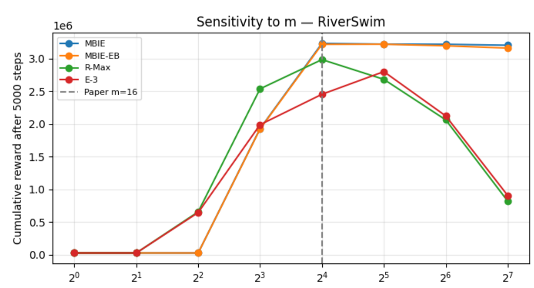

Figure 6: Sensitivity to m on RiverSwim. Dashed line marks the paper's value $m = 16$. Mean over 5 trials.

MBIE and MBIE-EB. On RiverSwim, both algorithms perform well for $m \geq 2^4 = 16$ and remain stable for larger

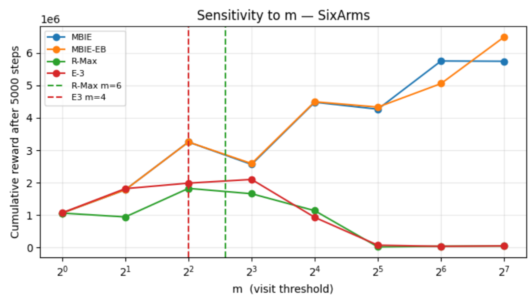

Figure 7: Sensitivity to m on SixArms. Dashed lines mark the paper's values: R-Max $m = 6$ (green) and E^3 $m = 4$ (red). Mean over 5 trials.

values (≈ 3.2M). For small m ($\leq 2^2 = 4$), performance degrades significantly as the model is frozen after too few observations. On SixArms, both algorithms continue to improve with larger m, reaching their best performance at $m = 2^7 = 128$ (MBIE: ≈ 5.7M, MBIE-EB: ≈ 6.5M). Their CI widths adapt automatically, making them far less sensitive to m than threshold-based methods.

R-Max and E^3. Both show a clear inverted U-shape on RiverSwim, peaking at $m = 2^4 = 16$ (the paper's value) and degrading on both sides: too small m yields an inaccurate model; too large m wastes the entire budget on exploration. On SixArms, the degradation for large m is more severe — R-Max and E^3 collapse to near-zero reward for $m \geq 2^5 = 32$, confirming the paper's values ($m = 6$ and $m = 4$ respectively) are carefully calibrated.

6.3. Extension 3: Sensitivity to A and B

MBIE has two parameters controlling confidence interval width: A for the reward CI and B for the transition CI. The paper uses $A = 0.3$, $B = 0$ for RiverSwim and $A = 0.3$, $B = 0.08$ for SixArms without systematic justification. We ran a grid search over $A \in \{0.1, 0.3, 0.5, 1.0, 2.0\}$ and $B \in \{0.0, 0.05, 0.08, 0.1, 0.3\}$ (5 trials, 5000 steps each) and report mean final cumulative reward as a heatmap (Figure 8).

RiverSwim. Performance decreases monotonically as A increases: a larger reward CI means more optimism, more exploration, and fewer steps left to exploit within 5,000 steps. The paper's setting ($A = 0.3$, $B = 0$, red box) lies in the high-performance region. Crucially, $B = 0$ (ignoring transition uncertainty entirely) performs as well as any other B value.

SixArms: The pattern is more complex. The paper's setting ($A = 0.3$, $B = 0.08$) is not optimal, better performance is achieved at $A = 0.3$, $B = 0.1$ (reward $\approx 11.7M$). The paper reports only that parameters were *found during a broad search*, without specifying the grid size or number of trials, which may partly explain the discrepancy between our optimal parameters and theirs. Unlike RiverSwim, SixArms is more sensitive to B, reflecting that transition uncertainty matters more when arms have very different success probabilities.

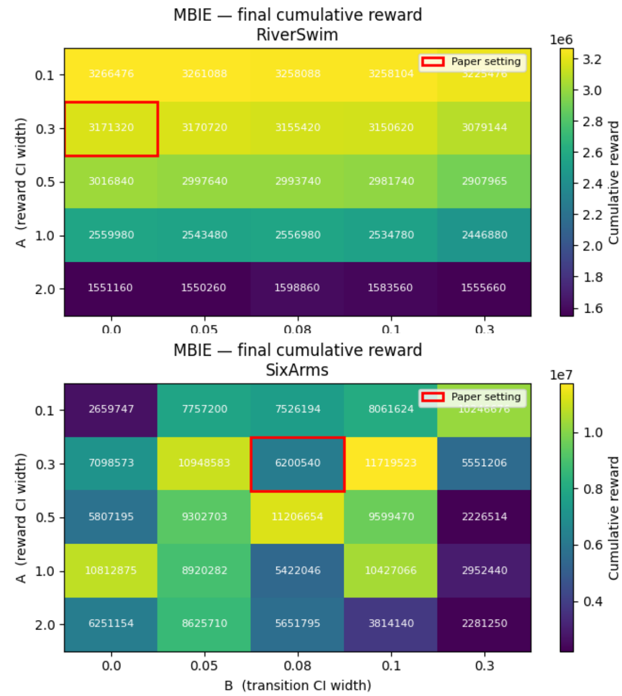

Figure 8: MBIE performance heatmap over $A \times B$ on RiverSwim (top) and SixArms (bottom). Red box marks the paper's setting. Mean over 10 trials.

6.4. Extension 4: Sensitivity to γ

The paper fixes $\gamma = 0.95$ throughout. The theoretical sample complexity scales as $1/(1-\gamma)^6$, meaning the bound is extraordinarily sensitive to γ in theory. We swept $\gamma \in \{0.5, 0.7, 0.8, 0.9, 0.95, 0.99\}$ across all four algorithms (5 trials, 5000 steps each).

RiverSwim (Figure 9). Increasing γ consistently improves performance for all algorithms. This is expected: RiverSwim requires long-term exploration, as the high reward is located at the far-right state and requires repeatedly swimming against the current. A larger γ encourages agents to value delayed rewards and persist in exploration. MBIE and MBIE-EB show the strongest improvement, reaching best performance around $\gamma = 0.95$-0.99. R-Max also improves but plateaus earlier around $\gamma = 0.8$, suggesting diminishing returns once optimism has already driven exploration toward the rewarding state. E3 stabilizes after $\gamma = 0.7$, reflecting that its explicit explore/exploit switching benefits less from

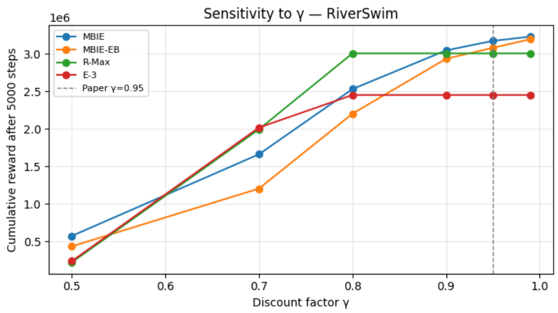

Figure 9: Sensitivity to γ on RiverSwim. Dashed line marks $\gamma = 0.95$. Mean over 10 trials.

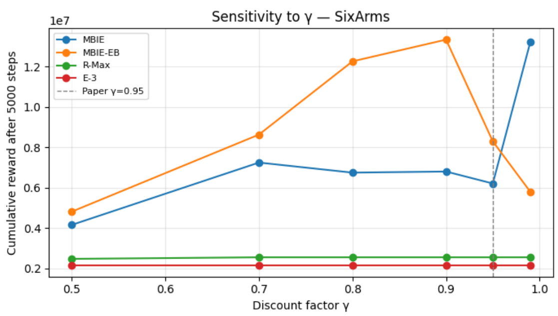

Figure 10: Sensitivity to γ on SixArms. Dashed line marks $\gamma = 0.95$. Mean over 10 trials.

longer planning horizons. Overall, the paper's choice of $\gamma = 0.95$ is well justified for RiverSwim.

SixArms (Figure 10). The effect of γ is more algorithm-dependent. MBIE-EB peaks at $\gamma = 0.9$ then drops at $\gamma = 0.95$, suggesting that excessive emphasis on future rewards leads to over-exploration. MBIE improves until $\gamma = 0.7$, becomes unstable, then reaches another peak at $\gamma = 0.99$, possibly benefiting from extremely long planning horizons, though this likely depends on A and B calibration. R-Max and E3 remain flat across all γ values: their performance is limited by their exploration mechanisms rather than the planning horizon. The paper's $\gamma = 0.95$ is reasonable for RiverSwim but not always optimal across all algorithms and environments.

6.5. Extension 5: FrozenLake 8×8 (Slippery)

FrozenLake 8×8 (OpenAI Gymnasium Towers et al. 2023) has 64 states, 4 actions, sparse reward (1.0 at goal, 0 elsewhere), and stochastic transitions: the intended action succeeds with prob. $1/3$, slips sideways with prob. $2/3$. Terminal states (holes and goal) auto-reset to the start state. We set $R_{\max} = 1/3$ (maximum expected reward per step),

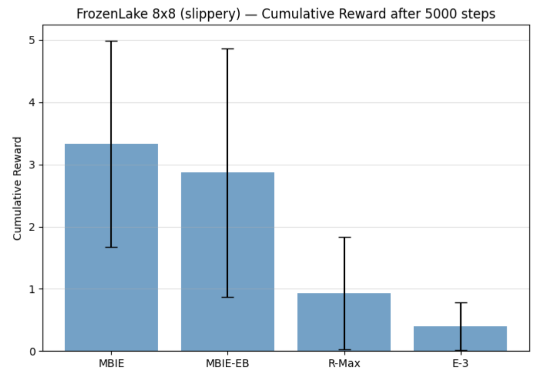

Figure 11: FrozenLake 8×8 (slippery) cumulative reward after 5,000 steps. $R_{\max} = 1/3$, $\gamma = 0.95$, 5 trials.

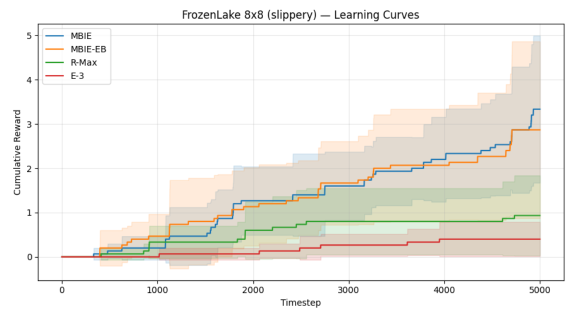

Figure 12: FrozenLake 8×8 (slippery) learning curves (mean $\pm$ std, 5 trials).

$\gamma = 0.95$, and $m = 2$ for R-Max and E^3.

Results. MBIE leads ($\approx 3.33 \pm 1.66$), followed by MBIE-EB ($\approx 2.87 \pm 2.00$), R-Max ($\approx 0.93 \pm 0.90$), and E^3 ($\approx 0.40 \pm 0.39$). Learning curves show a staircase pattern, each step representing a successful goal reached. MBIE and MBIE-EB begin accumulating reward from ≈ 500 steps; R-Max from $\approx$1,500 steps; E^3 barely accumulates reward throughout. With 256 state-action pairs and $m = 2$, at least 512 exploration steps are needed before any pair is known to R-Max — E^3's binary switch then causes it to spend nearly the entire remaining budget in exploration mode.

This extension confirms that the optimism principle generalises to standard RL benchmarks beyond the paper's designed environments, and that CI-based methods scale better to larger state spaces than threshold-based methods.

7. Discussion

Confidence interval vs. threshold methods. MBIE and MBIE-EB consistently outperform R-Max and E^3 across all experiments. The key advantage is *partial exploitation*: confidence interval widths shrink continuously with each visit, allowing exploitation to begin before a pair is fully "known". This is most pronounced on SixArms, where the $3\times$ gap reflects the advantage of ruling out low-reward arms early.

Sensitivity to m. MBIE and MBIE-EB are largely insensitive to m once $m \geq 16$, while R-Max and E^3 show a clear inverted U-shape — collapsing to near-zero reward for $m \geq 32$ on SixArms. The theoretically optimal $m \approx 200{,}000$ is four orders of magnitude larger than the empirically effective $m = 16$, illustrating that PAC-MDP bounds are conservative worst-case guarantees rather than practical recommendations.

Role of the transition confidence interval. The transition confidence interval (B parameter) has little effect on RiverSwim ($B = 0$ is optimal) but matters on SixArms, where arms have very different success probabilities. This explains why MBIE-EB sometimes matches MBIE and sometimes falls short.

Discount factor and generalisation. On RiverSwim, $\gamma = 0.95$ is well calibrated — all algorithms improve monotonically with γ. On SixArms, the optimal γ differs between algorithms, suggesting a non-trivial interaction between the exploration bonus and the planning horizon. Finally, Extension 5 confirms that the optimism principle generalises to FrozenLake 8×8, where MBIE and MBIE-EB collect over $3\times$ the reward of R-Max and E^3.

8. Conclusion

We reproduced the main experiments of Strehl and Littman (2008) and confirmed their central finding: confidence-interval methods outperform threshold-based methods because they begin exploiting an advantageous state-action pair as soon as sufficient evidence accumulates. On RiverSwim all four algorithms collect $\approx$3M reward, with MBIE leading slightly. On SixArms the gap is striking: MBIE-EB ($\approx$8.3M) and MBIE ($\approx$6.2M) collect more than three times the reward of R-Max ($\approx$2.5M) and E^3 ($\approx$2.1M).

Our five extensions show that confidence interval based methods are insensitive to the visit threshold m, transition faster from exploration to exploitation, and generalise to new environments such as FrozenLake 8×8. Threshold-based methods are sensitive to m and waste exploration steps on actions they have already identified as suboptimal.

One consistent gap between theory and practice is that the theoretically optimal $m \approx 200{,}000$ is four orders of magnitude larger than the empirically effective $m = 16$, illustrating that PAC-MDP bounds are conservative worst-case guarantees rather than practical recommendations.

Future work could explore extending these algorithms to MDPs with continuous or infinite state and action spaces, as suggested by the paper, and investigating tighter sample complexity bounds using KL-divergence rather than L_1-based confidence intervals.

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LLM Usage

Large language models were used to support this project in three ways: understanding and clarifying theoretical concepts from the paper, assisting with code debugging and implementation, and improving the clarity of the writing. The models used were ChatGPT (OpenAI, 2025), Claude (Anthropic, 2025), and Gemini (Google DeepMind, 2025). All mathematical content, experimental results, and scientific conclusions are our own.

A. GitHub Repository

Our full implementation is publicly available at:
https://github.com/allizha/mdp-interval-estimation-study